



Design, fabrication and test of a solid propellant microthruster array by conventional precision machining



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ABSTRACT

Prototypes of 10×10 and 100×100 scale solid propellant microthruster (SPM) array were fabricated with conventional precision machining, due to restriction on the manufacture of large-scale microthruster array with present MEMS manufacturing technique. The microthruster consists of ignition circuit layer, ignition powder layer, propellant layer, cavity layer and nozzle layer. Row–column driving control plan was applied in ignition circuit to realize simultaneous ignition and reduce the number of wires. The solid propellant was AP-HTPB pellet propellant with a diameter between $100 \mu\text{m}$ and $200 \mu\text{m}$. A test to choose proper ignition powder was performed among black powder, modified black powder and self-made ignition powder, which demonstrated the reliability of self-made ignition powder and realize 100% ignition success rate. In the mean time, a measurement of ignition resistor's surface temperature was taken to determine ignition delay, together with numerical simulation and ignition resistor fusing test. Thrust was also measured for the 10×10 scale microthruster array, and the results show that the specific impulse is much lower than calculated result, so a more comprehensive numerical model is in need to depict the working process of a single SPM. The test also demonstrates the feasibility of large-scale solid propellant microthruster array's fabrication by conventional fabrication methods.

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1. Introduction

Future space missions will rely on micro/nano spacecrafts to adjust attitude or orbits, so a propulsion system will be a necessary requisite [1–3]. Current propulsion systems are either too large, or need so much power that micro/nano spacecraft cannot afford. The micro/nano spacecrafts require a propulsion system to deliver low thrust values and impulse in short time [4]. Compared to other micro propulsion techniques, the solid propellant microthruster has advantages of simple structure, high reliability, no frictional forces inherent to moving parts, no leakage of propellant and high propellant stability [5–7]. The major disadvantages are its low specific impulse and one-shot use, which can be mitigated by adopting an array of multiple microthrusters fabricated on a baseplate.

Relevant techniques of SPM arrays have been systematically studied in US, France, Singapore, Japan, South Korea and Canada [8–11]. Whereas, most of their research was aimed at the fabrication and the test on small scale SPM arrays with MEMS techniques, of which the thrusters on each array were no more than 100. MEMS fabrication method is suitable for mass fabrication, but the proce-

dures of MEMS fabrication method is more complicated compared to conventional fabrication methods. Conventional fabrication methods such as laser drilling and mechanical drilling have advantages of fewer procedures and flexibility when re-design is in need. Another advantage of conventional fabrication is that the materials feasible for fabrication are varied, while only a few materials, like silicon, can be etched by applying MEMS techniques. Our work mainly focuses on the feasibility of large scale array fabrication, so we will use conventional methods to fabricate SPM arrays, which has been shown to be feasible in small scale SPM array fabrication [12]. Future space missions using micro/nano spacecrafts will need larger scale SPM arrays, so our work mainly focuses on fabrication, control system and the test of SPM arrays with 100 and 10,000 microthrusters to verify the performance of the SPM arrays and the feasibility of large scale SPM array fabrication with conventional methods.

2. Structure of the array

2.1. Structure of single thruster

A single SPM is composed of ignition circuit layer, ignition powder layer, propellant layer, cavity layer and nozzle layer, and the nozzle outlet is sealed with cover. Most SPM array proto-

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Nomenclature

φ_c	Pressure fluctuation coefficient
$P_{\max}$	Maximum pressure in combustion chamber
D_{tc}	Diameter of the combustion chamber
$[\sigma_{tc}]$	Allowable stress
$[\epsilon_{tc}]$	Allowable strain
E_{tc}	Young's modulus
ϵ_r	Surface emitting rate
σ_0	Stefan Boltzmann constant
T_r	Radiation temperature
T_b	Real temperature
T_a	Atmospheric temperature
ρ	Density of the ignition resistor's materials
c	Specific heat of the ignition resistor's materials
λ	Thermal conductivity of the ignition resistor's materials
a_p	Pre-exponential factor of burning rate pressure exponent
n_p	Burning rate pressure exponent
T_f	Flame temperature of propellant combustion
r_p^0	Propellant powder's initial radius
ρ_p	Propellant density
r_p	Propellant powder's radius
ρ_c	Gas density in combustion chamber
p_c	Pressure in combustion chamber
r_b	Burning rate
k	Specific heat ratio
M_p	Propellant mass in the combustion chamber of each SPM
S	Burning surface area of solid propellant pellets in each SPM
Γ	A single valued function about specific heat ratio
R_0	Gas Constant
M	Molecular weight of gas
c^*	Characteristic velocity

types developed by researchers in the past are fabricated with MEMS processing techniques. Each layer is fabricated with conventional precision machining. The processing precision lies between 0.01 mm and 0.1 mm, which meets the demand for punching a micro hole with a diameter between 0.3 mm and 1.5 mm. At first laser drilling was selected as the drilling method, but the ablation effect will burn the edge of the wall, as is shown in Fig. 1, so mechanical drilling was actually applied then for fear that the ablation effect may reduce the wall strength of a single SPM and affect bonding of layers.

By applying high precision mechanical drilling methods with the help of CNC, fabrication of more than 1000 micro holes is feasible with high efficiency in one-time processing. The unpunched layer is fixed by a fixture assembly, then a reference position is set in the computer. According to the reference position, the first hole is punched. The punched hole is then used as the next reference for the next hole to be punched, so SPMs and alignment holes can be punched on all layers. Combining digital control technology to position each hole on the plate precisely, each layer of the SPM array can be fabricated.

The structure of a single SPM is shown in Fig. 2.

2.2. Material selection

Materials that have been applied to fabricate a SPM array mainly consist of silicon, machinable ceramic and photosensitive glass.

In our research, we chose epoxy resin as the material to fabricate the SPM array. Epoxy resin has perfect heat insulation property, of which the disadvantage is low strength. Therefore, strength checking is essential in the process of design.

2.3. Ignition powder and propellant

The ignition device of the SPM is a resistor. Due to the tiny scale of the heating surface, proper ignition powder with high sensitivity and high specific energy is requisite. The burning rate of the ignition powder should also be high enough to afford adequate energy releasing rate to efficiently ignite the propellant. AP-HTPB solid propellant was chosen as the tested propellant.

2.4. Structure design of the array

On the demand of fabricating large scale array and due to the restriction of fabricating methods, a multi-layer structure and a bottom ignition plan are applied. Our research mainly focuses on the feasibility of large scale array fabrication with conventional methods, so we chose a simple straight nozzle to substitute for a laval nozzle. On the premise that the heat loss and viscous loss are neglected, the thrust coefficient of the straight nozzle is 1.

According to existing research findings, no undesirable ignitions of neighboring thrusters are observed during the working process of a single thruster, when the wall material is MACOR[®] ceramic with a thickness of about 1 mm [6]. So it is believed that due to the better heat insulation property of epoxy resin compared with MACOR[®] ceramic, undesirable ignition will not happen to our SPM array with the same wall thickness. Whereas, a strength checking is essential to make sure that our SPM's wall thickness is enough to withstand high pressure in the combustion chamber.

Designed value of pressure in normal solid rocket motor's combustion chamber should lie between 7 MPa and 20 MPa to maintain the combustion of propellant, so we chose a value in this range, 12 MPa to decide the wall thickness. The minimum wall thickness was solved according to Eqs. (1) and (2).

$$\delta_{\min} = \frac{\varphi_c P_{\max} D_{tc}}{2.3 [\sigma_{tc}] - \varphi_{tc} P_{\max}} \quad (1)$$

$$[\sigma_{tc}] = E_{tc} [\epsilon_{tc}] \quad (2)$$

Epoxy resin is an isotropic material, so E_{tc} is considered equal to its Young's modulus, whose value is 34.138 GPa. The value of $[\sigma_{tc}]$ is 34.138 MPa, with $[\epsilon_{tc}]$'s value set to 0.001. The value of φ_c is set to 1.2. The value of D_{tc} is 1 mm. Accordingly, the value of $\delta_{\min}$ is calculated to be 0.2246 mm. For safety and heat insulation reasons, the wall thickness was set to 1 mm, which means that the minimum distance between two neighboring SPMs is 1 mm.

3. Ignition system

The ignition system is used to control the thruster ignition and the ignition sequence, as is shown in Fig. 3.

3.1. Ignition circuit layer

Most relative research facilities have realized the ignition of a single SPM by separated ignition circuit. Whereas, it is difficult to apply the circuit layout to large scale situations. For instance, an array consisting 10,000 SPMs need 10,000 circuits, which is hard to fabricate. To achieve the ignition of a large scale SPM array and combine it with the row–column addressing control method, a circuit was designed. Ignition resistors with a resistance value of 2 Ω were used to heat ignition powders to ignition temperature, which were laid at the bottom of the ignition layer of an SPM array, so the

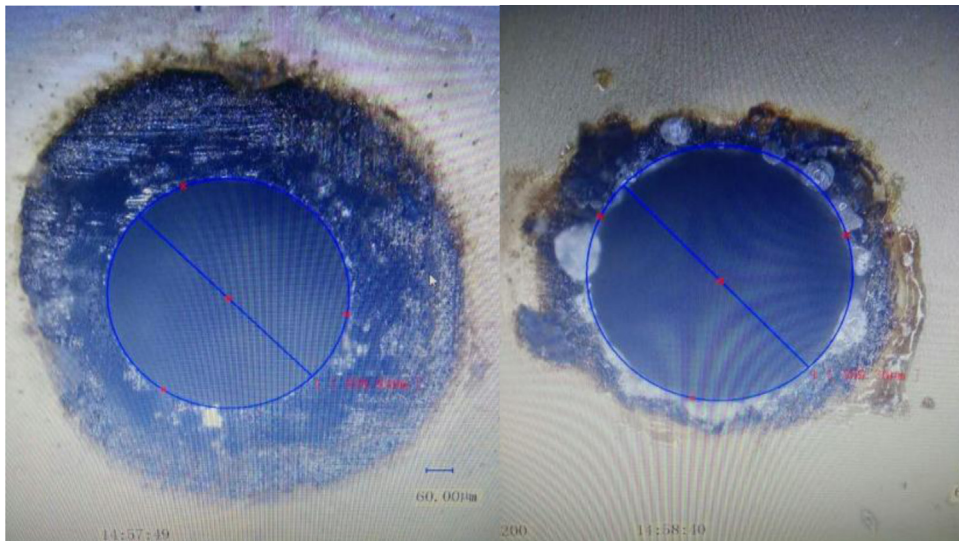


Fig. 1. Holes punched by laser drilling method.

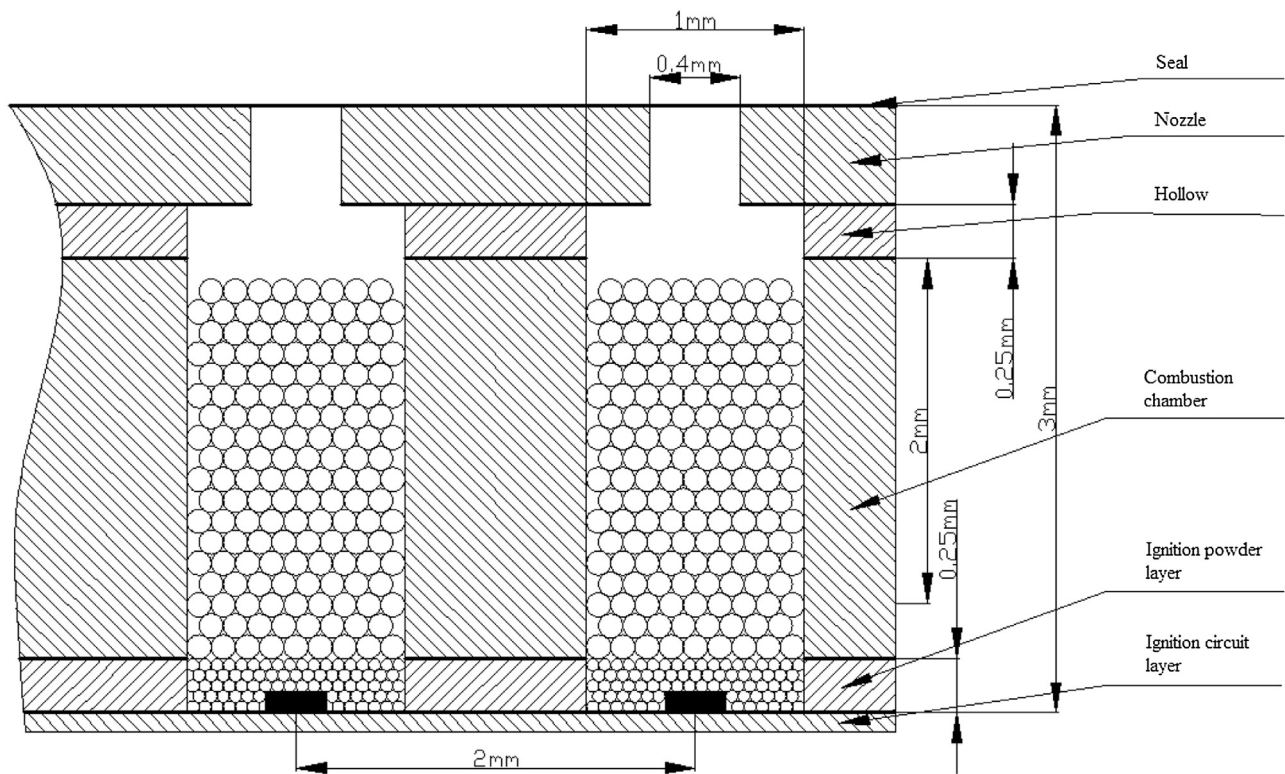


Fig. 2. A schematic illustration of a single SPM.

ignition circuit can be fabricated on a PCB board with conventional microelectronics processing techniques. Diodes are deposited on the other side of the PCB board and serial connected with ignition resistors to prevent undesirable ignition of SPMs not in need. According to Fig. 4, only 200 wires are available for ignition of any PM in an array consisting 10,000 SPMs. The manufactured ignition circuit is shown in Fig. 5.

3.2. Control circuit

The control circuit consists of a micro-control unit (MCU) and a drive circuit.

An FPGA chip was selected as the MCU of the ignition system, and the ignition sequence code was programmed in VHDL language. A USB port was used to communicate with the host computer.

3.3. Drive circuit

The drive circuit is composed of row driver and column driver, as is shown in Figs. 6–8. Considering significant ignition current, thyristors and Darlington power transistors are suitable for amplifying the current. Whereas, if both drivers are equipped with thyristors, only when the ignition resistor is burnt can the circuit be switched off, which is harmful to the controllability of ignition sequence. If both drivers are equipped with Darlington power tran-

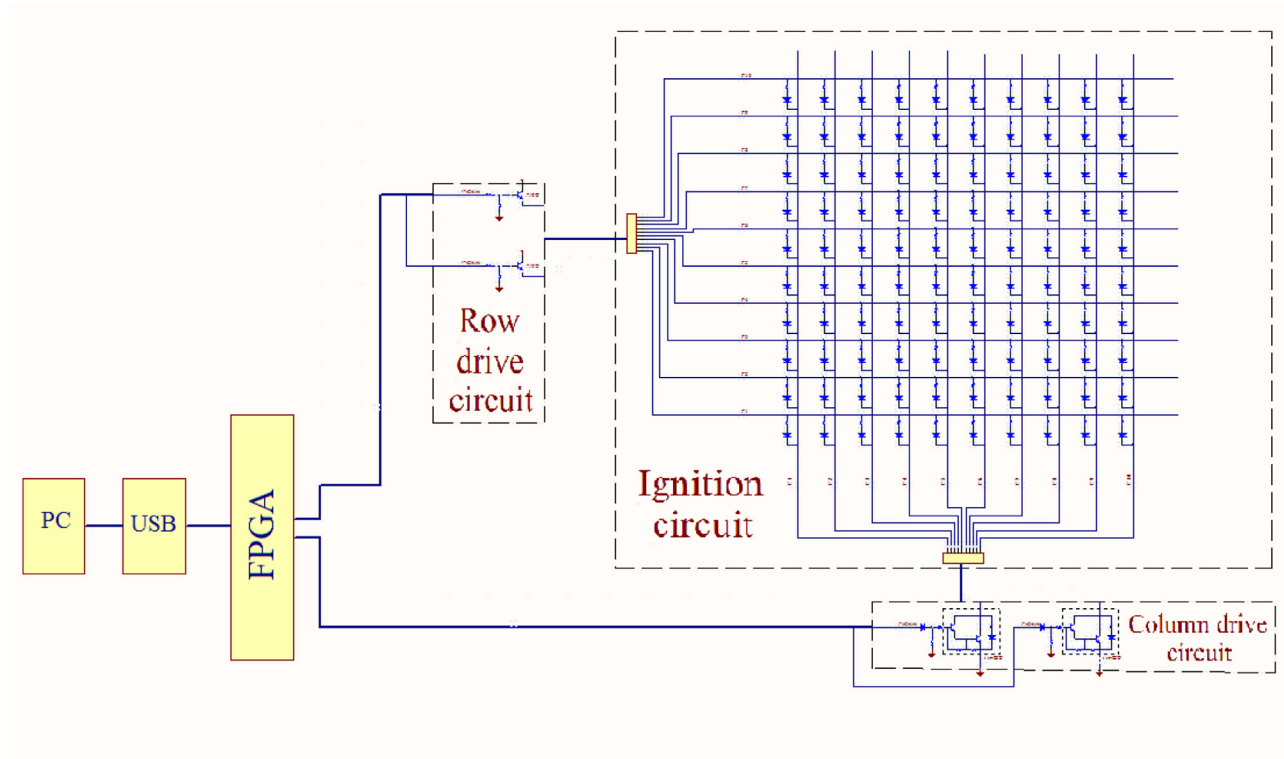


Fig. 3. A schematic illustration of the ignition system.

sistors, the circuit will become complicated. As a trade-off, the row driver is equipped with a Darlington power transistor, and the column driver is equipped with a thyristor. The thyristor is used as a switch, together with Darlington power transistors to perform row–column addressing method.

4. Assembly of the thruster array

Due to the multi-layer structure of the SPM array, each layer of the SPM array should be fabricated separately. Bonding process was implemented to assemble all parts.

The assembling procedures are shown below.

Grinding: Ignition powders and the propellant were grinded into particles with a grinder. Due to the danger in grinding ignition powders, various components were separately grinded. The components insensitive to grinding were directly grinded into small particles in the grinder, while the sensitive components were first manually grinded in large particles, then put into the grinder to grind into small particles in small batches, with the grinder working in lowest speed. Above work was done in an atmospheric temperature of 20 °C.

Screening: A pulp classifier was implemented to divide particles into different groups according to the diameter. Then, propellant particles with diameter between 100 μm and 200 μm were chosen to load the micro combustion chamber, while ignition powders were subdivided into two groups of diameter between 50 μm and 100 μm, and between 100 μm and 200 μm to make a comparison.

Loading: For it is hard to manually load particles into the SPM array, a vibrator was used to help load the ignition powders and propellant powders into combustion chambers by the effect of vibrating, as is shown in Fig. 9.

Weighing: A balance with an accuracy of 0.1 mg was used to weigh the mass of ignition powders and propellant particles loaded in each SPM. For we can not directly weigh the mass of particles

in each SPM, the mass here is the mean mass. For an SPM array containing n SPMs, the mass is calculated using Eq. (3):

$$\bar{m} = \frac{m_2 - m_1}{n} \quad (3)$$

where m_1 and m_2 are respectively the total masses before and after loading.

Bonding: Epoxy resin glue was used to join all layers together. Bolts were first put into alignment holes punched in the ignition circuit layer, which were located near the edge of the layer, and then fixed with the layer by using glue. With the help of bolts, ignition layer was aligned and bonded to the ignition circuit layer first, then loaded with ignition powders. The combustion chamber layer was then bonded to the ignition layer and loaded with propellant bonded to the assembled parts in the end.

Storage: After assembly, the SPM array was stored in an electric vacuum drying oven to keep ignition powders from moisture.

The ignition layer with ignition powders loaded and the assembled SPM array of 10 × 10 scale are shown in Figs. 10 and 11 respectively.

The ignition layer with ignition powders loaded and the assembled SPM array 100 × 100 scale are shown in Figs. 12 and 13 respectively.

5. Integral experiment of the thruster array

An integral experiment was carried out to test the reliability of the igniter, the feasibility of the conventional fabrication procedure and the performance of the SPM array. The prototype of 10 × 10 and 100 × 100 scale SPM arrays were tested in the experiment.

The equipments needed in the experiment are ignition system, prototype of a 10 × 10 scale SPM array, prototype of a 100 × 100 scale SPM array, a micro thrust stand and data acquisition system.

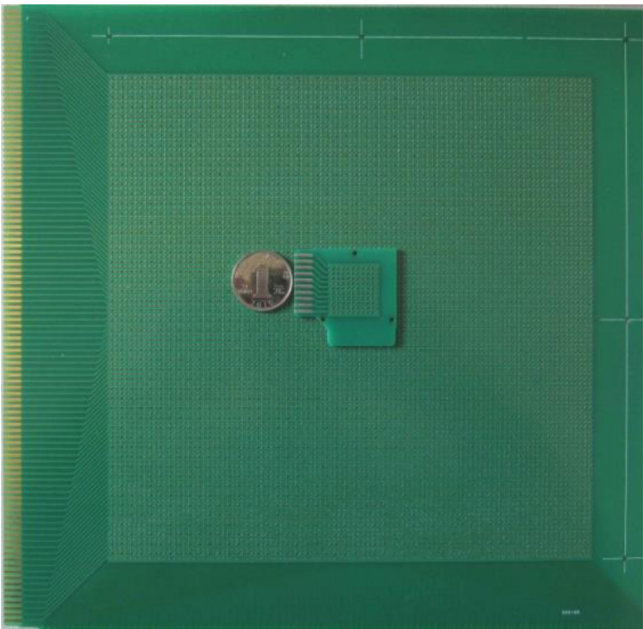


Fig. 5. Manufactured ignition circuit(for 10 × 10 and 100 × 100 array respectively).

5.1. Ignition device test

First, the ignition powders with higher ignition success rate were selected by ignition test. Then, the resistor fusing test and

infrared property test were performed to determine the range of ignition power. By numerical simulation, the time needed for the resistor surface temperature to reach the ignition powders' ignition point was determined.

For a single SPM in the array, the igniter is composed of ignition powders, the ignition layer and the ignition resistor. The success rate determines whether the SPM array can normally work, so an experiment was performed to test the heating behavior of the ignition resistor and the reliability of the igniter.

The 10 × 10 scale SPM array was chosen to test whether the igniter can normally work, so only the ignition layer was assembled. For each kind of alternative ignition powders, two ranges of particle diameter were tested. For each particle diameter, 30 thrusters were ignited, and the success rate was regarded as the criteria to select the proper kind of ignition powders. The test was performed with relatively higher power, of which the value was set to 1.62 W. The results are demonstrated in Table 1.

We can conclude from Table 1 that the igniter loaded with self-made ignition powders, which are mixture of black powders, red phosphorus and potassium chlorate, has higher ignition success rate. The black powders accounted for 50% of the total weight, while the ratio of red phosphorus and potassium chlorate were verified according to the solid-phase Reaction below. The components were put into a container in small batches after screening. Then, the components were manually mixed by shaking the container.

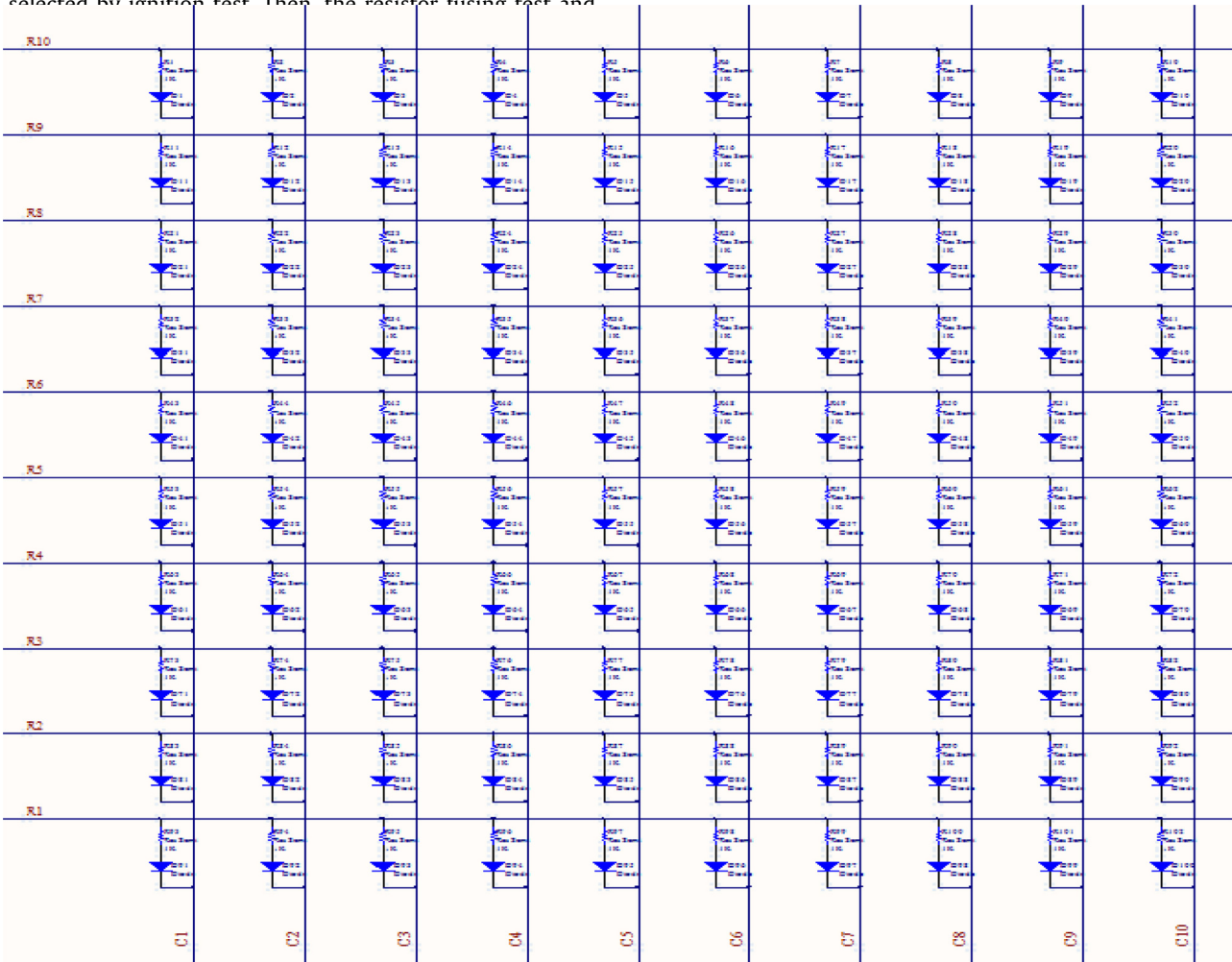
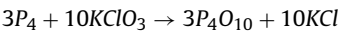


Fig. 4. A schematic illustration of the ignition circuit.

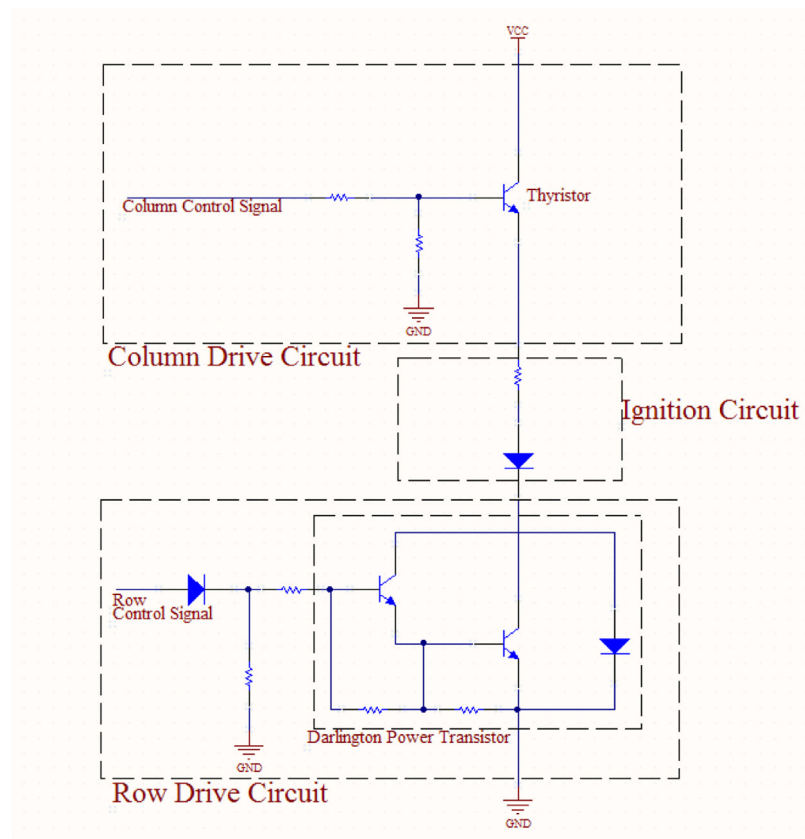


Fig. 6. A schematic illustration of single drive circuit.

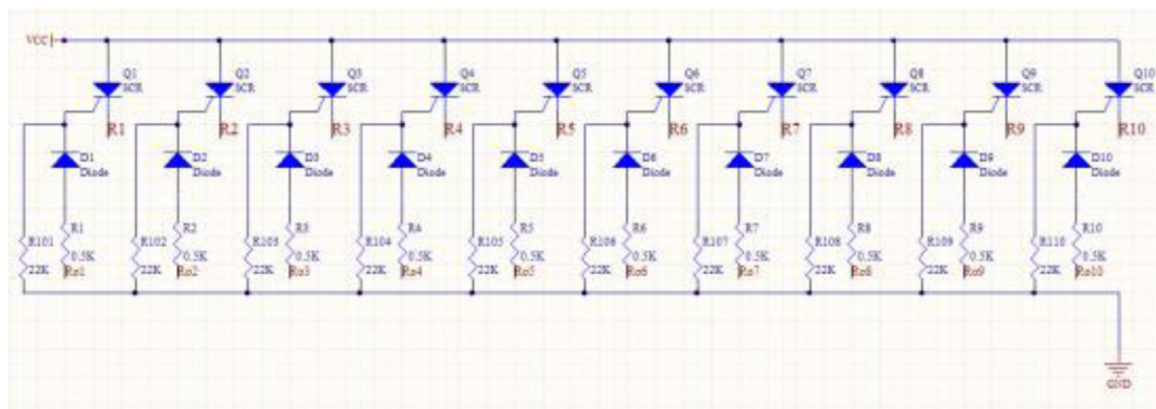


Fig. 7. A schematic illustration of column drive circuit.

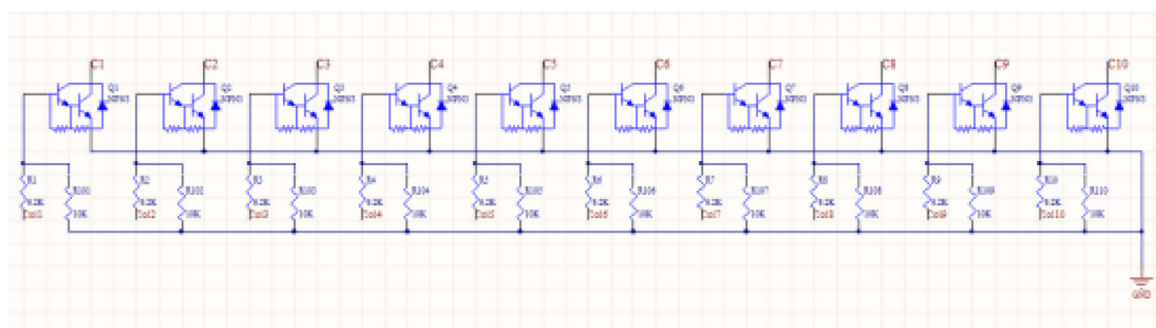


Fig. 8. A schematic illustration of row drive circuit for 10 × 10 array.



Fig. 9. The vibrator applied for loading the pellet propellant.

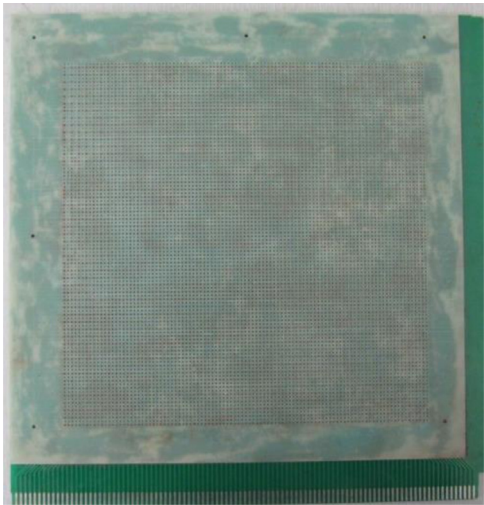


Fig. 12. Ignition layer of 100 × 100 thruster array.

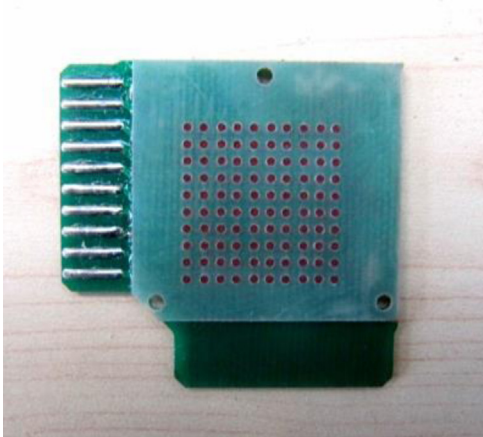


Fig. 10. Ignition layer of 10 × 10 thruster array.

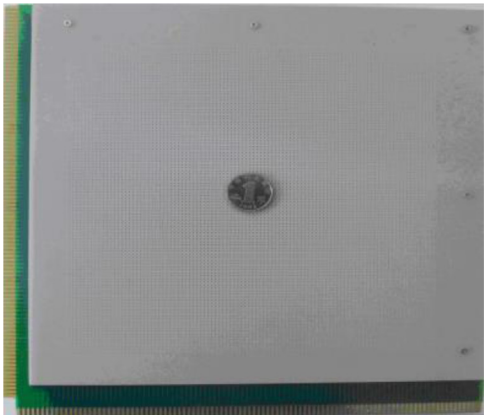


Fig. 13. Assembled 100 × 100 thruster array.

Table. 1
Ignition powders testing results.

Ignition powder	Diameter distribution	Ignition success rate	Mean mass of ignition powder/mg
Black powder	50–100 μm	23.3%	0.127
	100–200 μm	10.0%	0.094
Modified black powder	50–100 μm	60.3%	0.142
	100–200 μm	36.7%	0.121
Self-made ignition powder	50–100 μm	100.0%	0.116
	100–200 μm	80.0%	0.083

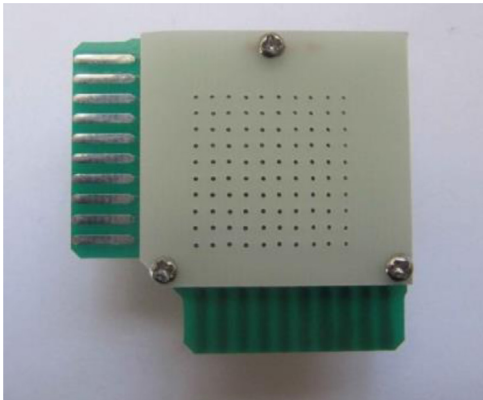


Fig. 11. Assembled 10 × 10 thruster array.

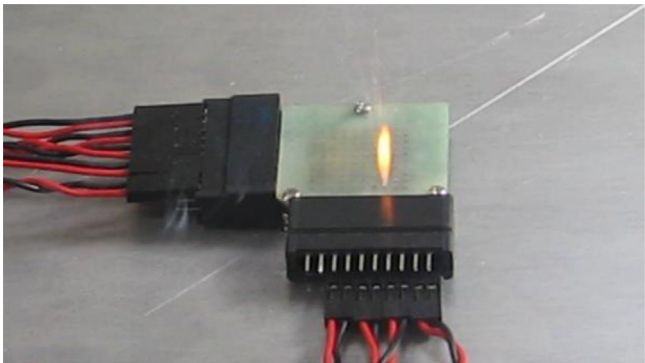


Fig. 14. Ignition transient of the igniter loaded with only self-made ignition powders.

We can also know that it is easier to successfully ignite the ignition powders of smaller diameter, for they have larger contact surface with the ignition resistor and larger burning surface to release more power compared with those with larger particle diameter [13]. Accordingly, the self-made ignition powders with particle diameter ranging from 50–100 μm were selected, and the ignition transient is shown in Fig. 14.

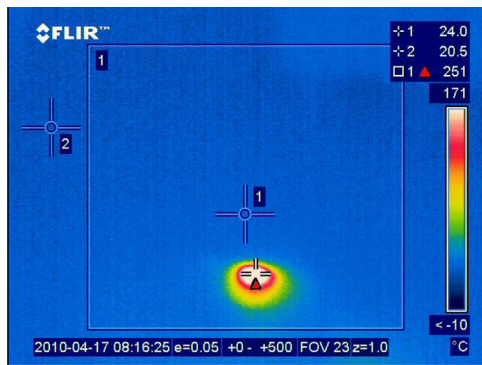


Fig. 15. Surface emitting temperature measured with thermal infrared imager at the moment of ignition.

5.2. Ignition resistor surface temperature measurement

A thermal infrared imager was used to measure the surface temperature distribution when the circuit was turned on. The surface

Table 2
Properties of materials.

Material	Density/kg × m ⁻³	Thermal conductivity/W × m ⁻¹ × K ⁻¹	Specific heat/J × kg ⁻¹ × K ⁻¹
Glass glaze	3800	35	778.2
Ruthenium oxide	6970	20.9	535
Ceramic	2650	11	960

emitting temperature was 251 °C at the moment of ignition, as is shown in Fig. 15.

$$T_b = \left[\frac{(\sigma_0 T_r^4 - (1 - \epsilon_r) \epsilon_r' \sigma_0 T_a^4)}{(\epsilon_r \sigma_0)} \right]^{1/4} \quad (4)$$

According to Eq. (4), the real surface temperature can be calculated, where ϵ_r is the emitting rate with a value ranging from 0.9148 to 0.9194. Therefore, the real temperature lies between 261.7 °C and 262.4 °C. Both values are beyond the ignition temperature of red phosphorus and potassium chlorate in self-made ignition powders and near the ignition point of black powders, which are major sources of gas products in self-made ignition powders. So, the temperature of 260 °C is regarded as the ignition temperature of ignition powders.

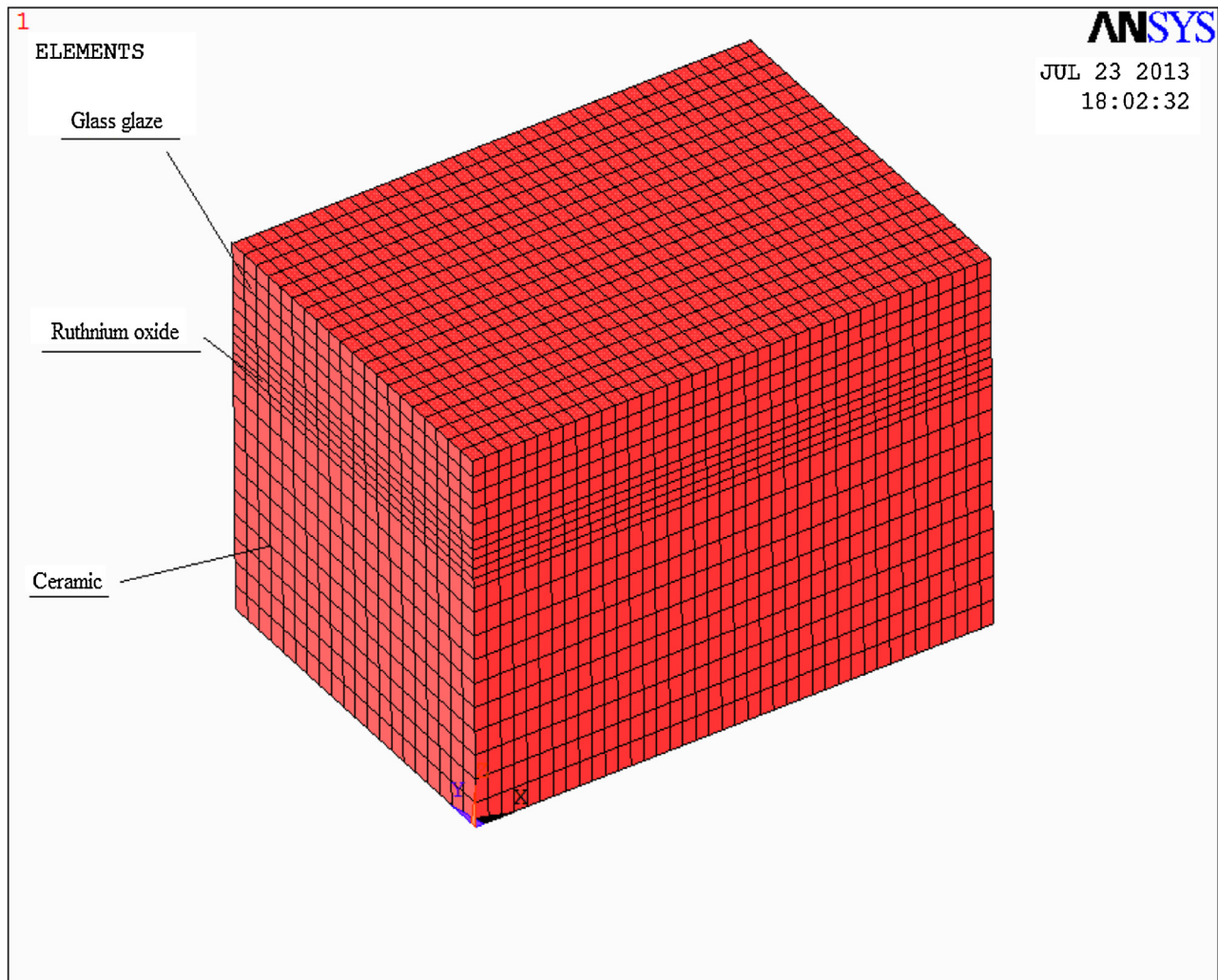


Fig. 16. Mesh of the resistor model.

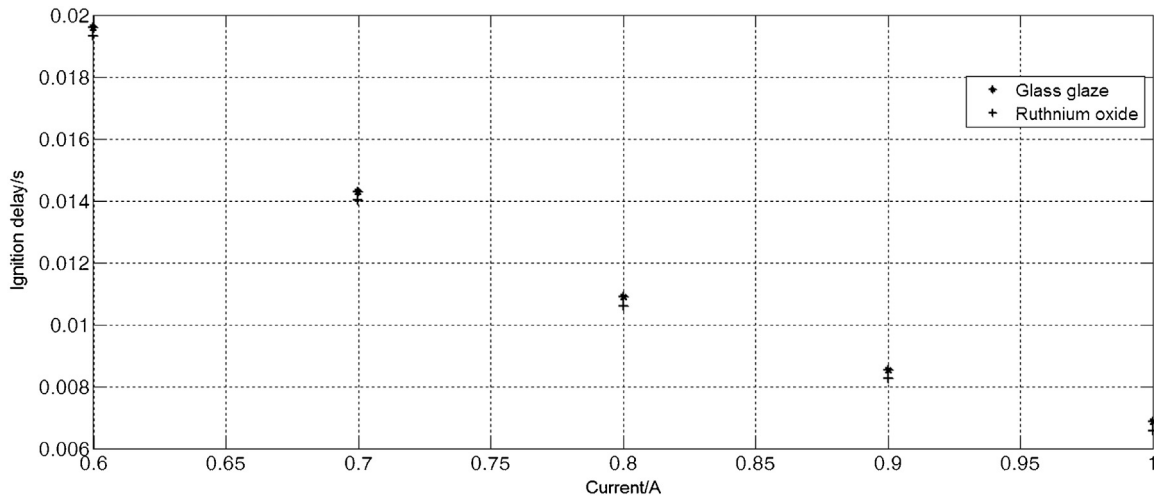


Fig. 17. Simulation results of ignition response time.

Accordingly, the heating property can meet the ignition demand, and the ignition delay will be determined by numerical simulation and relative experiment.

5.3. Test and simulation for ignition resistor heating process

The ignition resistor's fusing time was measured with various ignition power values. For each value of power, 15 thrusters were ignited. The time count began when the ignition resistor was heated, and ended until the resistor was fused.

Only the ignition powders were loaded in the test. According to the results, the lowest power available to ignite the ignition powder was 0.72 W, and the highest power required for a 100% ignition success rate was 2.65 W. The ignition success rate was 0% with a power of 3.38 W or larger, as is shown in Table 2.

Resistor heating process contains two phases, one is the heating effect of current, and the other is the heat conducted to the contact surface between the resistor and the ignition powder. We assume that the ignition powder is ignited as soon as the resistor surface temperature reaches ignition temperature in simulation. According to the heat transfer model, an unsteady heat transfer simulation was performed to calculate the time needed for the upper surface temperature of glass glaze and ruthenium dioxide to reach ignition point. The resistor variation during ignition was neglected due to the short duration and the low TCR coefficient with a value of about ± 100 ppm/°C. The simplified resistor model is shown in Fig. 16, the scale of the ruthenium dioxide layer is $0.6 \text{ mm} \times 0.3 \text{ mm} \times 0.03 \text{ mm}$, the scale of the glass glaze layer is $0.6 \text{ mm} \times 0.3 \text{ mm} \times 0.1 \text{ mm}$, and the scale of the ceramic layer is $0.6 \text{ mm} \times 0.3 \text{ mm} \times 0.2 \text{ mm}$, and the heat source was located in the ruthenium dioxide layer. The calculating mesh was hexahedral mesh. The properties of materials applied in the ignition resistor are shown in Table 2.

The unsteady heat transfer model was shown in Eq. (5):

$$\rho c \frac{\partial T}{\partial t} = \lambda \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) + \dot{Q}$$

For ruthenium dioxide layer, the value of heat source was equal to power generated by current, while for other layers the value of heat source was zero.

As is demonstrated in the simulation results, the ignition duration of the ignition powder diminishes as the ignition current, or the ignition power gets larger. The glass glaze layer's influence on the ignition duration before its upper surface temperature reaches ignition point, 260°C , is shown in Fig. 17. The duration values

Table 3
Ignition resistor fusing test.

Ignition power/W	Fusing time/ms	Success rate
0.72	>300	100%
0.98	270.7–272.4	100%
1.28	116.1–137.4	100%
1.62	81.3–87.7	100%
2	11.3–18.1	100%
2.65	5.7–7.9	100%
2.88	<4.5	60%

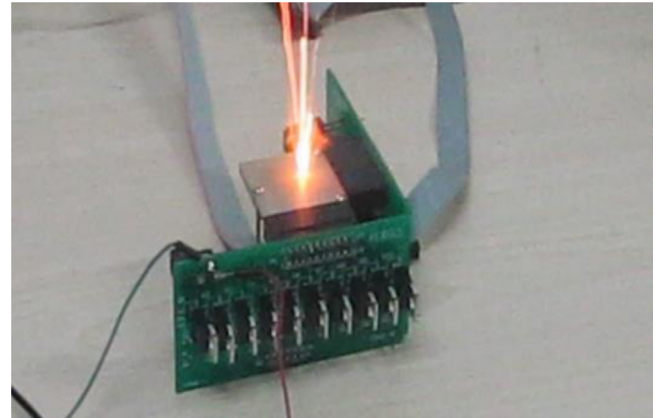


Fig. 18. Ignition transient of assembled 10×10 scale array.

corresponding to different ignition power were 19.6 ms, 14.3 ms, 10.9 ms, 8.5 ms and 6.8 ms respectively, which showed that the ignition was finished before the ignition resistor fused. The results of fusing test are shown in Table 3.

Whereas, when ignition power keeps increasing, the success rate will decrease. As is mentioned before, the resistor heating process contains both heating effect of current and heat conducted to ignition powders, so we deduce that the reason is that the ignition resistor fused too fast, even before the surface temperature of ignition powders could reach ignition point. In above simulation we did not take into consideration the heat conducted to ignition powders, and further study on this point will be conducted in the future.

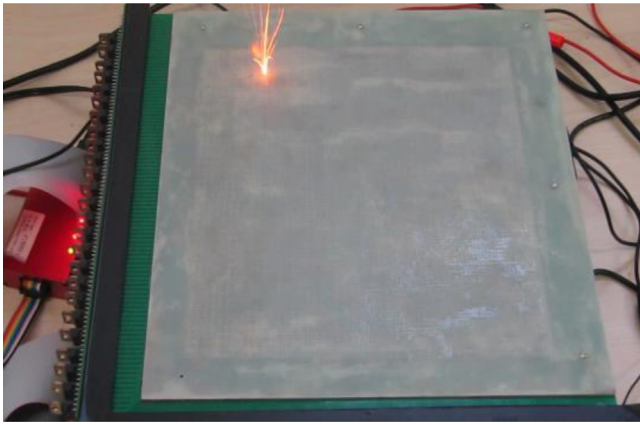


Fig. 19. Ignition transient of assembled 100 × 100 scale array.

5.4. Ignition test on totally assembled SPM arrays

The 10 × 10 scale array was tested first, with a power of 0.98 W and the duration was set 22 ms. To clearly distinct the successive ignition process, the interval was set 100 ms. Two 10 × 10 scale SPM arrays were tested with a success rate of 100%, the ignition transient is shown in Fig. 18.

The same set was implemented to the 100 × 100 array, and the success rate was 93.5%, with 650 SPMs not ignited. The main reason was that the adhesive surface was so large that clearance existed between the ignition layer and the ignition circuit, which was stuck by glue. The glue caused the ignition failure. The ignition transient is shown in Fig. 19.

5.5. Thrust measurement and relative analysis

The thrust measurement system shown in Fig. 20 was used to test the transient thrust of the solid propellant microthruster, of which the schematic was almost the same as that applied by Singapore National University [14]. The thrust was directly measured by a piezoelectric force sensor. The data acquisition system's sampling frequency was 500 KHz, and the piezoelectric quartz force sensor 208C01 manufactured by PCB company was chosen, of which the responding frequency ranges from 0.01 Hz to 36 KHz, with a maximum force threshold of 44.48 N. The ignition powders were self-made powders, and the propellant was AP-HTPB. The pro-

Table 5

Given parameters of AP-HTPB solid propellant.

$\rho_p / (\text{kg/m}^3)$	a_p	n_p	T_f / K	$M / (\text{kg/kmol})$	$r_p^0 / \mu\text{m}$
1670	0.004326	0.41	2500	23	75

(The unit of $a_p P^{n_p}$ is $\frac{\text{m}}{\text{s}}$, where P has a unit of Pa).

pellant mass was weighed according to the mass variation before and after loading, and then the mean mass of each thruster was obtained (Table 3 Fig. 20).

The 10 × 10 scale array was tested, with a mean loading mass of 0.103 mg for the ignition powder, and 0.928 mg for the propellant. The ignition power was 2 W. In total, 35 thrusters were ignited, of which the thrust variation was measured. The results are demonstrated in Figs. 21–24.

By thrust measurement, we can see a sequence of multiple thrust peaks with decreasing peak values. However, only one pulse was visually observed with our camera, so we can know that the process ended before the second peak occurred. It is believed that the emergence of subsequent peaks is mainly due to the oscillation of the thruster array and elastic deformation of the sensor itself.

During the whole process, no damage of any SPM was observed, so we are sure that SPM arrays fabricated with bonding process can meet the demand of structural strength. The specific impulse values in Table 4 are dispersed, which is mainly due to the non-uniformity in propellant diameter and propellant loading process of each thruster. The mean impulse was 117.67 N × s/kg.

“Thrust delay” is the time period from the time point at which the thrust value was larger than zero to the time point at which the impulse is equal to 3 μN × s, as Fig. 25 shows, which value is regarded as the least impulse to have effect on satellites' attitude and is set flexibly according to the requirement of satellite designers. The time period is usually used for statistics and analysis (Fig. 25).

6. Zero dimensional inner ballistic evaluation

A zero dimensional inner ballistic evaluation was performed to make a comparison with testing results.

The given parameters are shown in Table 5, and the seal opens when the pressure reaches 2.5 MPa.

Numerical model used in our experiments is described in Eqs. (6)–(23) [15]:

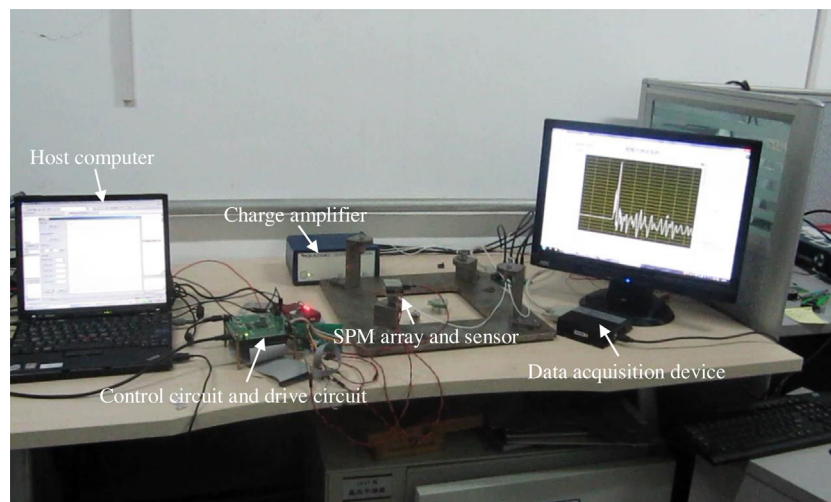


Fig. 20. Thrust stand and data acquisition system.

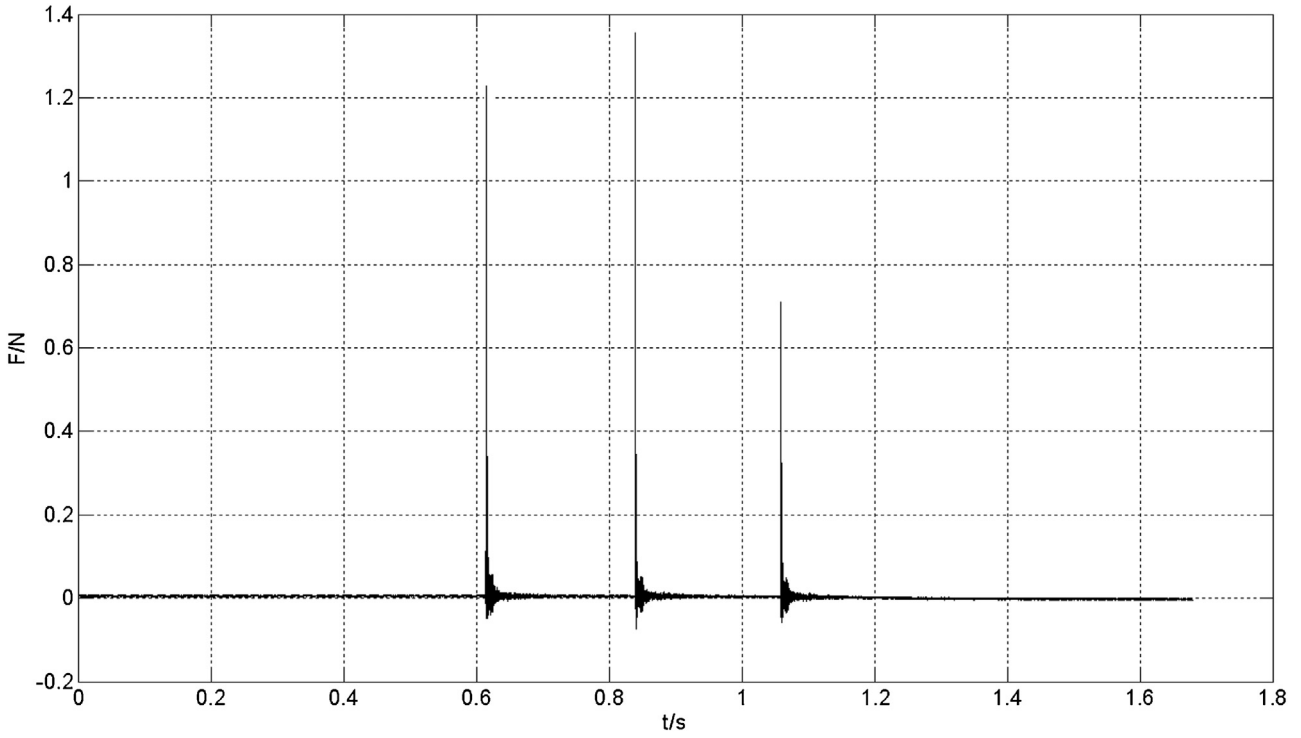


Fig. 21. Thrust-time curve of three successively ignited thrusters.

Before seal opening:

$$\frac{d\rho_c}{dt} = \frac{1}{V_p} \left(\rho_p S r_p - \rho_c \frac{dV_p}{dt} \right) \quad (6)$$

$$\frac{dp_c}{dt} = \frac{1}{V_p} \left(\rho_p S r_p k (\Gamma c^*)^2 - p_c \frac{dV_p}{dt} \right) \quad (7)$$

$$\frac{dV_p}{dt} = S r_b \quad (8)$$

$$\frac{dr_p}{dt} = r_b = -a_p \left(\frac{p_c}{10^6} \right)^{n_p} \quad (9)$$

$$r_i = 0 \quad (10)$$

After seal opening:

$$\frac{d\rho_c}{dt} = \frac{1}{V_p} \left(\rho_p S r_p - \rho_c \frac{dV_p}{dt} - \Gamma \frac{p_c A_t}{\sqrt{p_c/\rho_c}} \right) \quad (11)$$

$$\frac{dp_c}{dt} = \frac{1}{V_p} \left(\rho_p S r_p k (\Gamma c^*)^2 - p_c \frac{dV_p}{dt} - \Gamma p_c A_t k \sqrt{p_c/\rho_c} \right) \quad (12)$$

$$\frac{dV_p}{dt} = S r_b \quad (13)$$

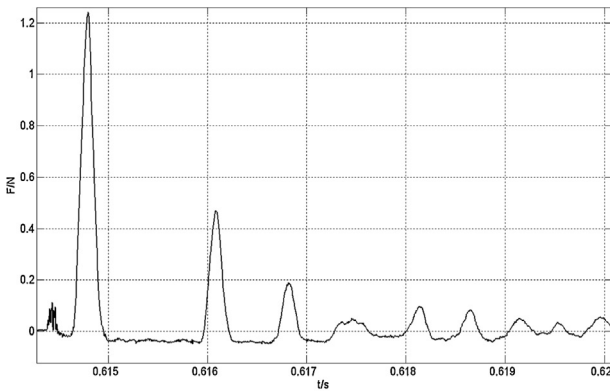
$$\frac{dr_p}{dt} = r_b = -a_p \left(\frac{p_c}{10^6} \right)^{n_p} \quad (14)$$

$$r_i = 0 \quad (15)$$

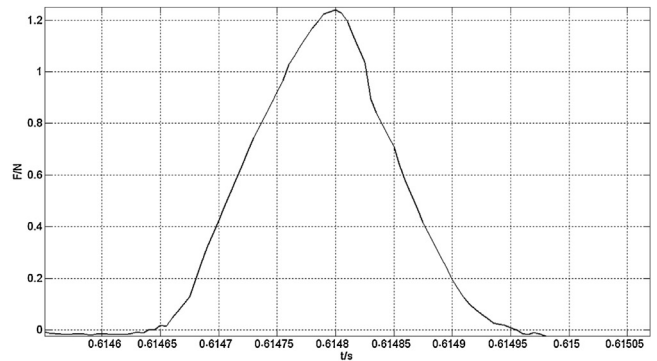
After combustion ending:

$$\frac{d\rho_c}{dt} = \frac{1}{V_p} \left(-\phi \Gamma \frac{p_c A_t}{\sqrt{p_c/\rho_c}} \right) \quad (16)$$

$$\frac{dp_c}{dt} = \frac{1}{V_p} \left(-\phi \Gamma p_c A_t k \sqrt{p_c/\rho_c} \right) \quad (17)$$

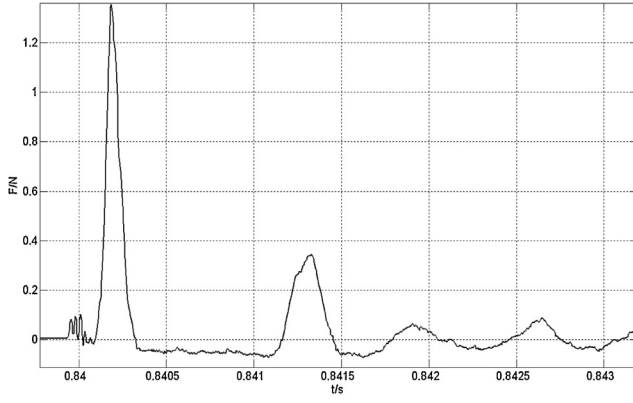


(a) Thrust-time curve.

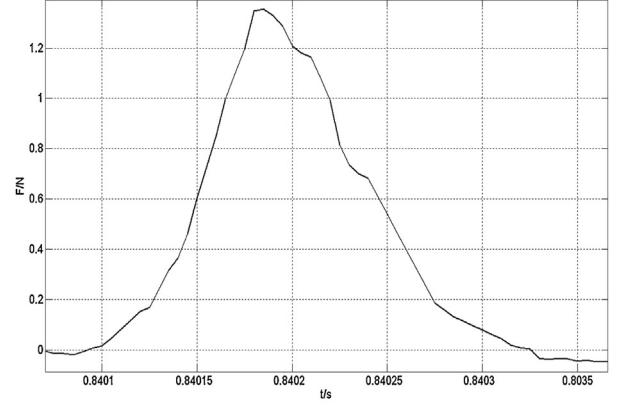


(b) Effective thrust-time curve.

Fig. 22. Thrust-time curve of the first thruster ignited in Fig. 21.

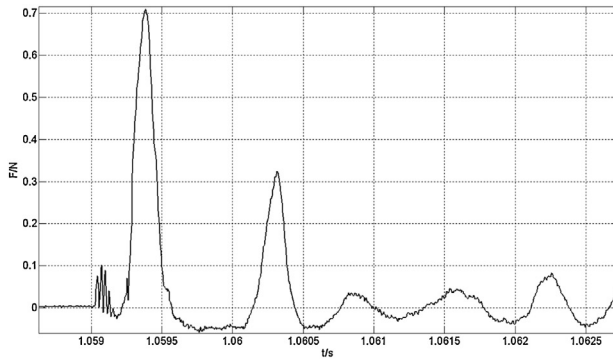


(a) Thrust-time curve.

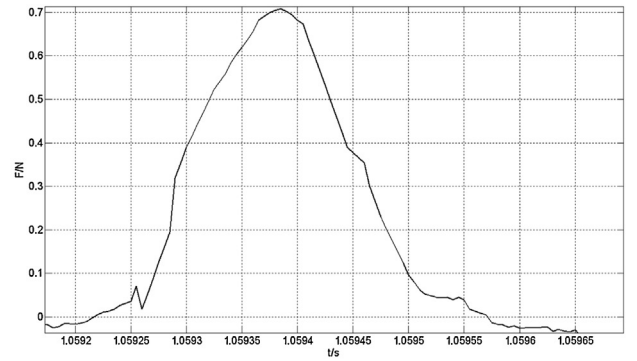


(b) Effective thrust-time curve.

Fig. 23. Thrust-time curve of the second thruster ignited in Fig. 21.



(a) Thrust-time curve.



(b) Effective thrust-time curve.

Fig. 24. Thrust-time curve of the third thruster ignited in Fig. 21.

Table 6
Calculation results.

Thrust peak value/N	Thrust duration/ms	Total impulse/mN × s	Specific impulse/N × s/kg
0.37	15	1.3705	1477

$$\frac{dV_p}{dt} = 0 \quad (18)$$

where

$$A_t = \pi r_t^2 \quad (19)$$

$$\Gamma = \sqrt{k} \left(\frac{2}{k+1} \right)^{\frac{k+1}{2(k-1)}} \quad (20)$$

$$c^* = \frac{\sqrt{R_0 T_f / M}}{\Gamma} \quad (21)$$

$$S = M_p / \frac{4}{3} \cdot 4\pi r_p^2 = \frac{3M_p}{r_p} \quad (22)$$

thrust calculation: For a straight nozzle, $C_F = 1$, so

$$F = C_F p_c A_t = p_c A_t \quad (23)$$

calculated thrust-time curve is shown in Fig. 26.

The results in Table 6 show that the peak thrust value was only 0.37N, much less than the measured value, while the thrust duration was more than 15 ms, much more than measured ones, which were less than 1 ms. So a more accurate and comprehensive

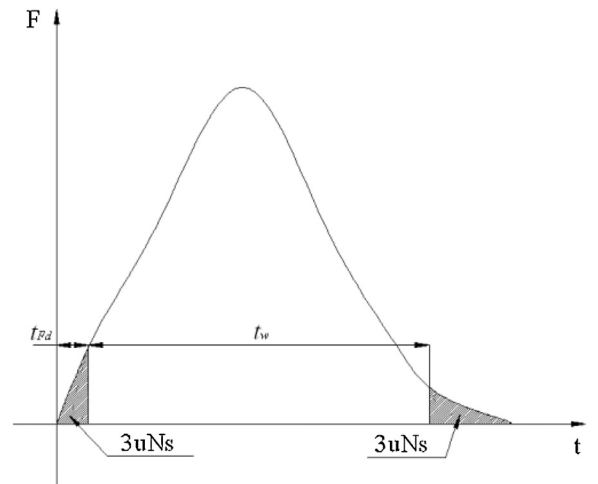


Fig. 25. Definition of thrust delay.

numerical model is required to describe the working process of the thruster (Table 6).

7. Conclusion

The design, fabrication and test of a solid propellant microthruster have been described in this paper. The conven-

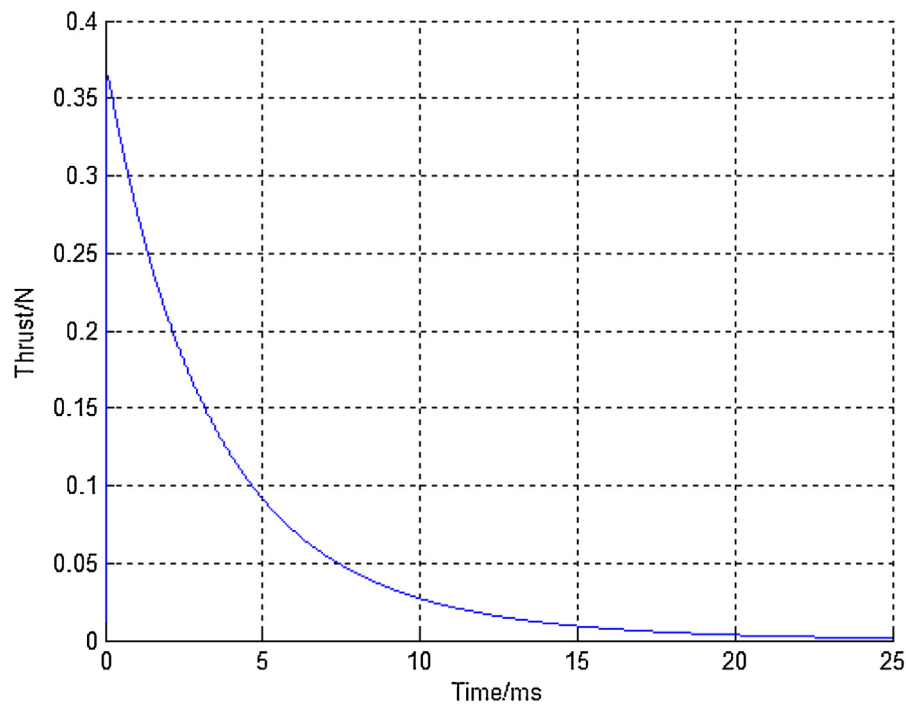


Fig. 26. Calculated thrust-time curve.

tional processing was chosen to fabricate the solid propellant microthruster arrays, rather than the MEMS technique. A 10×10 scale SPM array and a 100×100 scale array were fabricated, which contains 100 microthrusters and 10,000 microthrusters respec-

tively. The SPM array was fabricated in multi-layer structure. The row-column addressing layout was applied to ignition circuit to reduce the wire in requisite, which is appropriate for manufacture in large scale. The main components of the drive circuit are

Table 4
Thrust performance testing results.

Thruster number	Total impulse/mN $\times$ s	Working time/ms	Thrust delay/ms	Mean propellant mass/mg	Specific impulse/N $\times$ s/kg
1	0.1403	0.2668	0.0139	0.928	151.19
2	0.1394	0.2416	0.0127	0.928	150.22
3	0.0935	0.4086	0.0279	0.928	100.75
4	0.0909	0.3135	0.0238	0.928	97.95
5	0.077	0.3879	0.0301	0.928	82.97
6	0.1165	0.2431	0.0125	0.928	125.54
7	0.0805	0.4219	0.0200	0.928	86.75
8	0.1105	0.2956	0.0215	0.928	119.07
9	0.1203	0.2212	0.0271	0.928	129.63
10	0.1297	0.2595	0.0264	0.928	139.76
11	0.1341	0.2063	0.0251	0.928	144.50
12	0.1163	0.3014	0.0214	0.928	125.32
13	0.0971	0.3472	0.0232	0.928	104.63
14	0.1161	0.2635	0.0125	0.928	125.11
15	0.2125	0.1548	0.0211	0.928	228.99
16	0.1221	0.3015	0.0209	0.928	131.57
17	0.1135	0.3167	0.0247	0.928	122.31
18	0.0795	0.3795	0.0371	0.928	85.67
19	0.1167	0.3206	0.0245	0.928	125.75
20	0.0698	0.5887	0.0325	0.928	75.22
21	0.1272	0.2503	0.0287	0.928	137.07
22	0.1113	0.2737	0.0202	0.928	119.94
23	0.0973	0.3127	0.0153	0.928	104.85
24	0.0893	0.4605	0.0293	0.928	96.23
25	0.084	0.3947	0.0277	0.928	90.52
26	0.0808	0.4856	0.0296	0.928	87.07
27	0.1237	0.3339	0.0167	0.928	133.30
28	0.094	0.3435	0.0232	0.928	101.29
29	0.1255	0.3214	0.0291	0.928	135.24
30	0.1476	0.3101	0.0111	0.928	159.05
31	0.0966	0.4156	0.0257	0.928	104.09
32	0.0707	0.3732	0.0211	0.928	76.19
33	0.1306	0.277	0.0135	0.928	140.73
34	0.1084	0.3896	0.0177	0.928	116.81
35	0.0586	0.4616	0.0456	0.928	63.15

thyristors and Darlington power transistors, which can perform as a switch while amplifying the current needed to ignite the ignition powder. FPGA was chosen as the main control chip for fast processing speed and the ability to communicate with the host computer through USB port. To reduce heat loss, epoxy resin was chosen as the wall material. The ignition powders and propellants were grinded in pellet form to facilitate ignition process, and to broaden effective burning face to gain proper pressure for propellant combustion, as well as to adapt the head ignition layout. By testing three types of ignition powders with various diameters, the self-made powders with a diameter between 50 μm and 100 μm were chosen as the ignition powders. The propellant was grinded HTPB propellant powders with a diameter between 100 μm and 200 μm . The ignition temperature was measured 261.7 °C by using a thermal infrared imager, and the result was confirmed by transient heat transfer numerical simulation. The lowest power to reach a total ignition success was 0.72 W. The thrust and impulse experiments were performed to measure the performance of the 10 $\times$ 10 array, and the mean specific impulse was 117.67 N $\times$ s/kg. The ignition success rate experiments were performed to testify the pressure bearing ability of the combustion wall and to verify the feasibility of conventional fabrication method in large scale array fabrication. The thrust-time curve of a single thruster was acquired by solving a series equations, but the results were not align with the measured values, so a more comprehensive numerical model is required to depict the thruster's working process. The fabrication and experiments on 100 $\times$ 100 array demonstrated the feasibility of large scale SPM array fabrication.

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